Project Progress Tracker

Tim

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Contents

2026-03-19 **2**
Task 1: Concept Erasure Papers . . . 2
SPEED (ICLR'26 Poster) . . . 3
HiRM (ICLR'26 Poster) . . . 3
EraseAnything (ICML'25) . . . 3
Prototype-Guided Concept Erasure (CVPR'26) . . . 3
EraseAnything++ (arXiv'25) . . . 3
DVE – Differential Vector Erasure (ICML'26 under review) . . . 4
SURE – Subcategorical Erasure (ICLR'26 under review) . . . 4
Task 2: COCO Score Calculation . . . 4
Task 3: Defense Methods Evaluation . . . 4
Task 4: SPEED on SD 3.5 (Try 1) . . . 7
Architecture Mapping . . . 7
Key Issue: Weight Delta Too Large . . . 7
Best Result (scale=0.3 + retain set) . . . 7
Takeaway . . . 7

2026-03-20 **9**
Task 1: Choose Defense Methods Based on Paradigm . . . 9
Task 2: Recheck Existing Benchmarks . . . 9

2026-03-21 **10**
Task 1: What methods do existing papers compare? How do they tell their stories? . 10
Task 2: Rethink a Better Way to Tell Attack Story . . . 10
Task 3: Show other methods do not work on one case in our method . . . 10
Task 4: Show our method works on the same case . . . 10
Task 5: . . . 10

2026-03-19

Task 1: Concept Erasure Papers

Goal: investigate how reviewers evaluate concept erasure methods on OpenReview, focusing on the two ICLR'26 papers cited in our related work (SPEED, HiRM). Look for any discussion of compositional attacks or structural limitations.

Table 1: Recent concept erasure methods and their evaluation models.

Paper	Eval Model	OpenReview Weaknesses
SPEED [ICLR'26, 8/6/6/6]	SD 1.4	(1) Erasure not thorough—CLIP scores higher than RECE/MACE, residual patterns remain (2) Robustness: paraphrased prompts can recover erased concepts (3) Background changes in output images after concept removal
HiRM [ICLR'26, 4/4/6/4]	SD 1.4	(1) Simple extension of RMU to diffusion models (2) Text encoder modification easily bypassed by swapping encoder (3) Limited multi-concept scalability (4) Small gains over Diff-Q; underperforms on adversarial attacks (UnlearnDiffAtk, MMA)
EraseAnything [ICML'25]	FLUX.1-dev	N/A (not on OpenReview)
TRACE [ICLR'26 rejected, 6/6/4/4]	SD 3.5, FLUX, Infinity-2B/8B, SD 1.5	(1) No NSFW evaluation despite being key motivation (2) No failure mode discussion for overlapping concepts (3) FID insufficient—should include HPSv3/Aesthetics (4) Limited baselines, missing training-based methods (5) Computational cost not reported
Diversified Unlearning [ICLR'26 under review, 2/6/6/2]	SD 1.4	(1) Only SD 1.4, no generalization to SD 3.5/FLUX (2) Marginal preservation improvements (3) No comparison with adversarial unlearning methods (AdvUnlearn, RACE, RECE) (4) ChatGPT-generated prompts—hallucination risk, unclear how many needed (5) Embedding vector sum may destroy semantics
Prototype-Guided [CVPR'26]	SD 1.4, SDXL, SD 3.5	N/A
~~EraseAnything~~ [arXiv'25]	Flux.1-dev, Open-Sora	N/A (arXiv preprint)
DVE [ICML'26 under review]	Flux.1-dev	N/A (arXiv preprint)
SURE [ICLR'26 under review, 2/4/4/2]	SD 1.4	(1) Lipschitz regularization lacks theoretical justification (2) Only tested on similar subcategories, unclear for diverse categories (3) Three of four loss components borrowed from prior work (4) Concept space limited to 100 CLIP words—may miss fine-grained subcategories
PromptGuard [NeurIPS'25]	SD 1.4	N/A

SPEED (ICLR'26 Poster)

- **Title:** SPEED: Scalable, Precise, and Efficient Concept Erasure for Diffusion Models
- **Authors:** Ouxiang Li, Yuan Wang, Xinting Hu, et al.
- **OpenReview:** `https://openreview.net/forum?id=aoEtzdRkGh`
- **Scores:** 8, 6, 6, 6
- Method: Null space, SVD, erased concept mapping to existing concepts.

HiRM (ICLR'26 Poster)

- **Title:** Localized Concept Erasure in Text-to-Image Diffusion Models via High-Level Representation Misdirection
- **Authors:** Uichan Lee, Jeonghyeon Kim, Sangheum Hwang
- **OpenReview:** `https://openreview.net/forum?id=JSjOlFLUsS`

EraseAnything (ICML'25)

- **Title:** EraseAnything: Enabling Concept Erasure in Rectified Flow Transformers
- **Authors:** Daiheng Gao et al.
- **arXiv:** `https://arxiv.org/abs/2412.20413`
- **Architecture:** FLUX (rectified flow transformer / MMDiT). Paper claims SD3 support but code is FLUX-only.
- Method: Bi-level optimization with LoRA on `add_q_proj`/`add_k_proj`. Lower level erases target concept (ESD loss + attention regularizer); upper level preserves unrelated concepts (LoRA reconstruction + reverse self-contrastive loss). 3.57 MB per concept.
- **Note:** Only existing concept erasure work on flow-based T2I models. Could potentially be adapted to SD 3.5 (same MMDiT architecture).

Prototype-Guided Concept Erasure (CVPR'26)

- **Title:** Prototype-Guided Concept Erasure in Diffusion Models
- **arXiv:** `https://arxiv.org/abs/2603.08271`
- **Eval:** SD 1.4 (main), SDXL, SD 3.5 (compatibility test).
- Method: Training-free. k-means clustering on concept embeddings to form prototypes; steers generation away from prototypes using near-miss pairs. Supports multi-concept erasure and broad categories (I2P).
- **Note:** One of few training-free methods tested on SD 3.5. Reports Ring-A-Bell/P4D/UnDiff ASR on SD 3.5.

EraseAnything++ (arXiv'25)

- **Title:** EraseAnything++: Enabling Concept Erasure in Rectified Flow Transformers Leveraging Multi-Object Optimization
- **arXiv:** `https://arxiv.org/abs/2603.00978`
- **Eval:** Flux.1-dev (image), Open-Sora (video).
- Method: Extension of EraseAnything with multi-objective optimization (Pareto), reverse self-contrastive loss, and volumetric attention regularization for video. LoRA-based on MMDiT.
- **Note:** First concept erasure work on video diffusion models. Also tested entity/abstraction/relationship categories.

DVE – Differential Vector Erasure (ICML'26 under review)

- **Title:** Differential Vector Erasure: Unified Training-Free Concept Erasure for Flow Matching Models
- **arXiv:** `https://arxiv.org/abs/2602.01089`
- **Eval:** Flux.1-dev.
- Method: Training-free. Computes differential velocity vector between anchor and erase concepts, projects out harmful direction from generation trajectory. No fine-tuning needed.
- **Note:** Best ASR on Flux across I2P/MMA/RaB. Training-free approach achieving SOTA on flow-matching models.

SURE – Subcategorical Erasure (ICLR'26 under review)

- **Title:** Achieving Subcategorical Erasure in Text-to-Image Models
- **OpenReview:** `https://openreview.net/forum?id=1K7TSEe2x5`
- **Scores:** 2, 4, 4, 2
- **Eval:** SD 1.4.
- Method: Lipschitz regularization to ensure smooth model behavior around target category, enabling erasure of all subcategories using only the parent category label. Neutral eraser + preservation objective.
- **Note:** Addresses subcategory erasure problem (erase "guns" erases all gun types). Only SD 1.4.

Task 2: COCO Score Calculation

Goal: generate 1,000 images from MS-COCO captions (following SPEED's protocol) using SD 1.4 and SD 3.5-Medium, then compute CLIP Score to measure text–image alignment.

- Captions: first 1,000 from `mscoco.csv` (COCO val2014, sourced from SPEED repo).
- CLIP model: `openai/clip-vit-large-patch14`.
- SD 1.4: 20 inference steps, guidance scale 7.5.
- SD 3.5-Medium: 28 inference steps, guidance scale 7.5.

Table 2: CLIP Score ($\times 100$) on 1,000 COCO captions.

Model	CLIP Score	FID
SD 1.4	26.61	–
SD 1.4 (SPEED)	26.53	–
SD 3.5-Medium	27.03	–

Task 3: Defense Methods Evaluation

Goal: demonstrate that existing defense methods fail on compositional attacks (CompHarm benchmark, SD 3.5-Medium).

Reproduced SPEED's instance erasure on SD 1.4, erasing "Snoopy". Left: original SD 1.4; Right: after SPEED erasure.

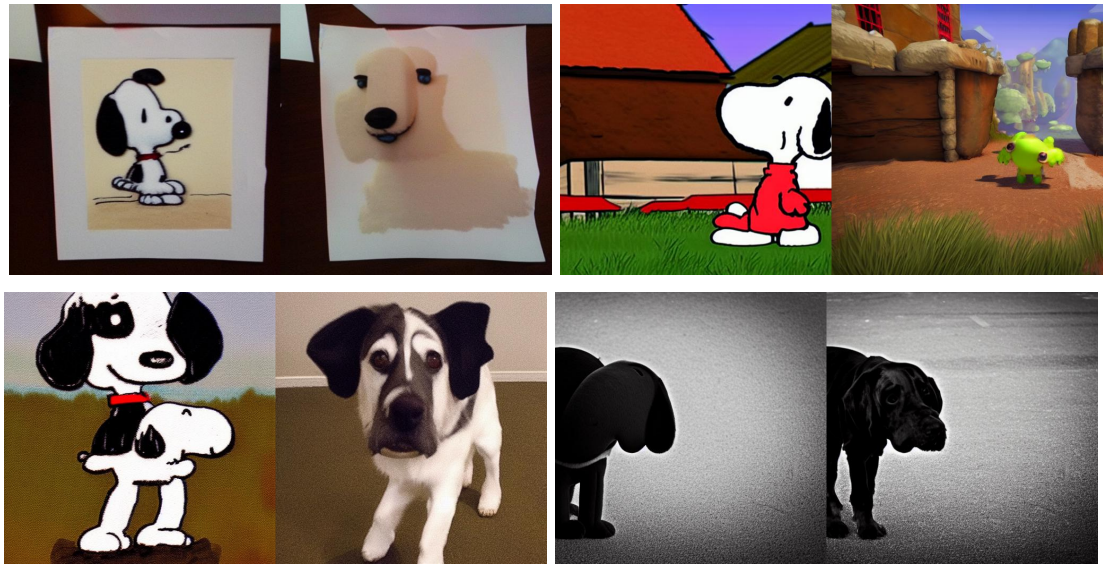

Figure 1: SPEED erasure of Snoopy on SD 1.4. Left: original; Right: after erasure.

Table 3: Existing defense methods to evaluate. ASR (%) detected by NudeNet (threshold 0.6), lower is better. Results from original papers.

Paradigm	**Method**	**Eval Model**	**I2P↓**	**MMA↓**	**RaB↓**	**CLIP↑**	**FID↓**	**Note**
Training-free guidance	SAFREE [ICLR'25]	SD 1.4	–	–	–	–	–	
	SafeDenoiser [NeurIPS'25]	SD 1.4	–	–	–	–	–	
	STG [NeurIPS'25]	SD 1.4	–	–	–	–	–	
Concept erasure	MACE [CVPR'24]	SD 1.4	1.19	1.60	2.11	0.293	12.32	from HiRM T3
	SPEED [ICLR'26]	SD 1.4	0.20	0.24	0.20	0.263	37.82	w/o AT; T9
	SPEED w/ AT [ICLR'26]	SD 1.4	0.10	0.01	0.00	0.260	39.51	w/ AT; T9
	HiRM-S [ICLR'26]	SD 1.4	0.66	3.30	0.00	0.306	6.75	K-77; T3
	HiRM-R [ICLR'26]	Flux	3.23	9.70	44.21	0.308	–	K-77; T4
	TRACE [rejected]	SD 3.5	–	37.21†	25.21†	–	–	†style erasure robustness, not nudity
	TRACE [rejected]	Flux	–	18.89†	15.62†	–	–	†style erasure robustness, not nudity
	Div. Unlearn [rejected]	SD 1.4	*172	–	38.95	0.310	14.21	*I2P NudeNet total count; RaB K16 ASR; best variant (Div-AP)
	EraseAnything [ICML'25]	Flux	4.70	11.20	58.90	0.302	21.75	RaB Ring16; from DVE T2
	EraseAnything+Flux [arXiv'25]	Flux	*182	–	–	0.304	21.67	*I2P NudeNet total; TII
	DVE [ICML'26 under review]	Flux	2.40	7.10	5.30	0.300	21.70	Training-free; RaB Ring16; T2
	Prototype-Guided [CVPR'26]	SD 1.4	‡5.2	–	6.70	–	–	‡I2P Q16 overall; RaB ASR; T1-2
	Prototype-Guided [CVPR'26]	SD 3.5	–	–	31.00	–	–	Training-free; RaB ASR; T3
	SURE [under review]	SD 1.4	§1.70	–	–	–	14.02	§NER-0.3; T2
Prompt filt.	SafeGuider [CCS'25]	SD 1.4	–	–	–	–	–	

Task 4: SPEED on SD 3.5 (Try 1)

Goal: adapt SPEED's concept erasure from SD 1.4 (UNet) to SD 3.5-Medium (MMDiT).

Architecture Mapping

	SD 1.4	**SD 3.5**
Backbone	UNet	MMDiT (24 transformer blocks)
Edited layers	`attn2.to_k/v`	`attn.add_k/v_proj`
Text encoder	CLIP (768-d)	CLIP-L + CLIP-G + T5 → 4096-d
Embedding dim	768	4096 $\xrightarrow{\texttt{context_embedder}}$ 1536

Key Issue: Weight Delta Too Large

With default SPEED hyperparams, the null space covers >98% of the 1536-d embedding space (vs. 768-d in SD 1.4), causing destructive weight updates. Added `edit_scale` parameter to control delta magnitude.

Config	$\Delta W/W$	**Result**
Default (scale=1.0, no retain)	4.2×	Noise output
scale=1.0, retain=instance.csv	2.5×	Still broken
scale=0.3, retain=instance.csv	0.75×	Image OK but erasure messy
scale=0.1, retain=instance.csv	0.25×	Image OK but Snoopy preserved

Best Result (scale=0.3 + retain set)

Snoopy identity is disrupted but output becomes incoherent — not a clean erasure.

Figure 2: SPEED on SD 3.5 (scale=0.3, retain=instance.csv). Left: original; Right: after erasure. Erasure disrupts images rather than cleanly removing the concept.

Takeaway

SPEED's null-space projection does not transfer well to MMDiT. The joint attention architecture responds differently to selective weight editing than UNet cross-attention. Possible next steps:

- Edit only later transformer blocks (preserve structural layers).
- Edit `context_embedder` (global 4096→1536 projection) instead of per-block layers.
- Adapt EraseAnything's LoRA-based approach (designed for MMDiT).

2026-03-20

Task 1: Choose Defense Methods Based on Paradigm

We categorize existing T2I safety defenses into three paradigms:

Table 4: Defense paradigms and representative methods.

Paradigm	Description	Methods
Input-level	Intercept or rewrite the prompt before it enters the model.	SafeGuider (SD 1.4), Latent Guard (SD 1.4), PromptGuard (SD 1.4)
Inference-time guidance	Steer the denoising process at inference without modifying model weights.	SAFREE (SD 1.4, SDXL), SafeDenoiser (SD 1.4), STG (SD 1.4), Prototype-Guided (SD 1.4, SDXL, SD 3.5), DVE (Flux)
Weight editing	Permanently modify model parameters to erase target concepts.	MACE (SD 1.4), HiRM (SD 1.4), SPEED (SD 1.4), ESD (SD 1.4), TRACE (SD 1.5, SD 3.5, Flux), EraseAnything (Flux)

Key argument: all three paradigms operate at the token/concept level and lack reasoning about entity relationships. Compositional intent attacks — where each token is benign but the combination is harmful — bypass all of them.

Task 2: Recheck Existing Benchmarks

Consolidated safety benchmark results from original papers. ASR (%) = fraction of generated images deemed unsafe, lower is better.

Table 5: Comprehensive safety benchmark results of defense methods (from original papers).

Paradigm	Method	Model	I2P↓	P4D↓	RaB↓	MMA↓	UnDiff↓	CLIP↑	FID↓
Input-level	SafeGuider	SD 1.4	5.48	0.46	0.01	2.05	–	0.275	–
	Latent Guard	SD 1.4	–	–	–	–	–	–	–
	PromptGuard	SD 1.4	–	–	–	5.53	–	0.260	–
Inf.-time guidance	SAFREE	SD 1.4	3.40	38.40	11.40	58.50	28.20	0.311	36.35
	SAFREE	SDXL	–	28.50	24.10	16.90	24.60	0.278	–
	SAFREE	SD 3	–	27.10	43.00	16.50	30.20	0.316	–
	SafeDenoiser	SD 1.4	–	–	12.70	46.90	20.70	0.307	22.55
	STG	SD 1.4	–	–	12.00	–	–	0.311	22.00
	Proto-Guided	SD 1.4	5.20	14.50	6.70	–	13.30	–	45.10
	Proto-Guided	SD 3.5	–	9.00	31.00	–	8.00	–	–
	DVE	Flux	2.40	–	5.30	7.10	–	0.300	21.70
Weight editing	ESD	SD 1.4	–	78.80	51.30	–	77.50	–	33.30
	MACE	SD 1.4	1.19	–	2.11	1.60	–	0.293	12.32
	SPEED	SD 1.4	0.20	–	0.20	0.24	–	0.263	37.82
	SPEED w/ AT	SD 1.4	0.10	–	0.00	0.01	–	0.260	39.51
	HiRM-S	SD 1.4	0.66	–	0.00	3.30	–	0.306	6.75
	HiRM-R	Flux	3.23	–	44.21	9.70	–	0.308	–
	TRACE	SD 3.5	–	–	25.21	37.21	–	–	–
	TRACE	Flux	–	–	15.62	18.89	–	–	–
	EraseAnything	Flux	4.70	–	58.90	11.20	–	0.302	21.75

Observations:

- Weight editing methods (SPEED, HiRM) achieve lowest ASR but at the cost of CLIP/FID degradation.
- Inference-time methods (SAFREE, Prototype-Guided) preserve quality better but have higher ASR, especially on adversarial attacks (MMA, P4D).
- SafeGuider achieves strong results across all attacks with minimal quality loss — but only tested on SD 1.4.
- Few methods are evaluated on SD 3.5 or Flux; cross-model generalization is largely untested.
- **None of these benchmarks test compositional intent attacks.**

2026-03-21

Task 1: What methods do existing papers compare? How do they tell their stories?

Don't need to understand everything. Focus on how they tell the story.

Task 2: Rethink a Better Way to Tell Attack Story

Task 3: Show other methods do not work on one case in our method

For example, IED. 5 examples about explosive devices.

Task 4: Show our method works on the same case

Task 5: